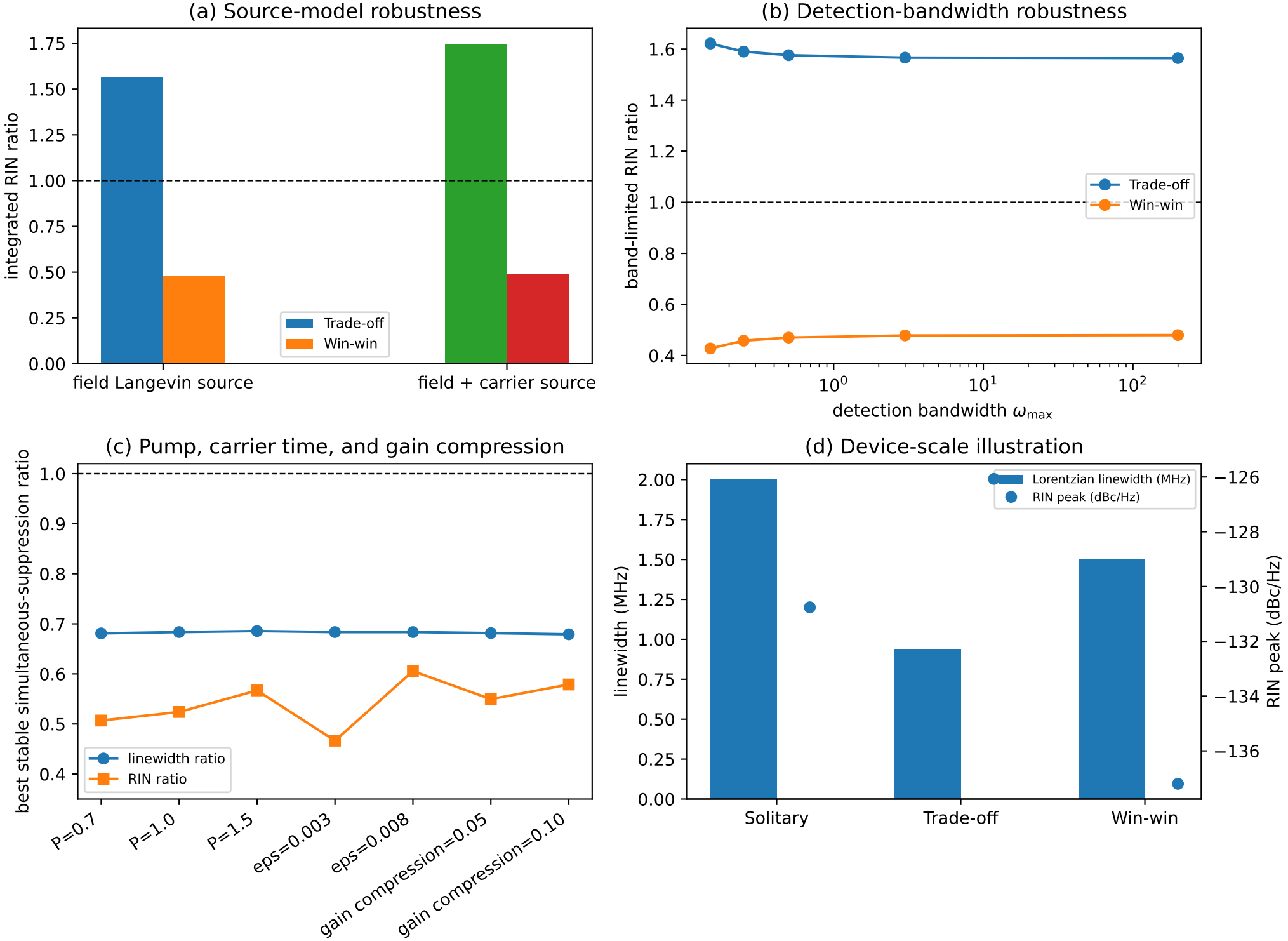
(a) Source-model robustness
integrated RIN ratio
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Trade-off
Win-win
field Langevin source
field + carrier source
(b) Detection-bandwidth robustness
band-limited RIN ratio
1.6
1.4
1.2
1.0
0.8
0.6
0.4
Trade-off
Win-win
10^0
10^1
10^2
detection bandwidth ω_{max}
(c) Pump, carrier time, and gain compression
best stable simultaneous-suppression ratio
1.0
0.9
0.8
0.7
0.6
0.5
0.4
linewidth ratio
RIN ratio
P=0.7
P=1.0
P=1.5
eps=0.003
eps=0.008
gain compression=0.05
gain compression=0.10
(d) Device-scale illustration
linewidth (MHz)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Lorentzian linewidth (MHz)
RIN peak (dBc/Hz)
−126
−128
−130
−132
−134
−136
RIN peak (dBc/Hz)
Solitary
Trade-off
Win-win